MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NOR, DIGITAL, KEY CODE EMP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NOR gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0609 failures per million operating hours for class I or 0.3230 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EMP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/263-1 for class I and M28787/263-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

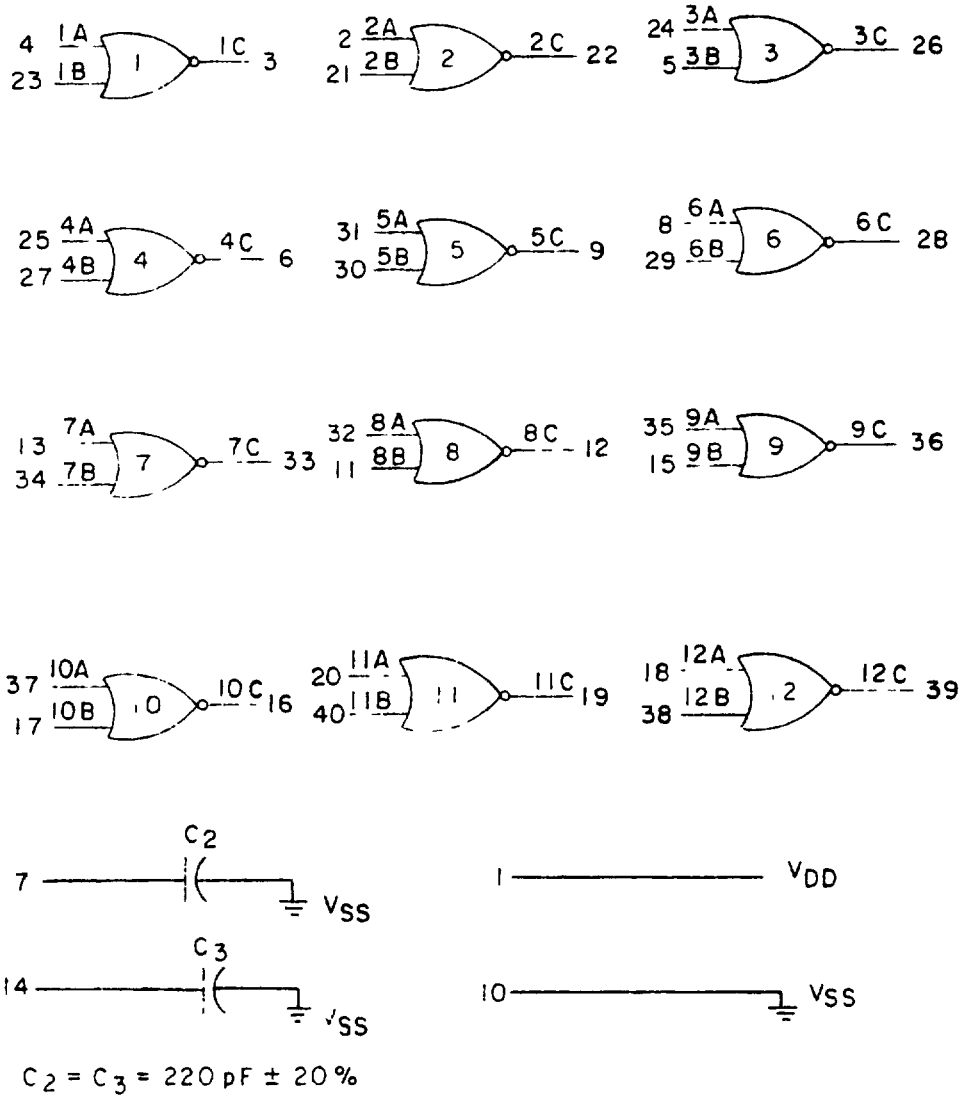
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



$C_2 = C_3 = 220\ pF \pm 20\%$

FIGURE 1. EMP module function diagram, NOR gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													-
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage (negative)	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage (positive)	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Output voltage (high level), I_{OH}= 0	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E12	Output voltage (low level), I_{OL}= 0	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F12	Output voltage (high level), V_{DD}= 5V	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G24	Output voltage (low level), V_{DD}= 5V	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H12	Output voltage (high level), V_{DD}= 15V	X	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
J1-J12	Output voltage, (low level), V_{DD}= 15V	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K24	Input current (high level)	XII	X	0	200	0	200	0	200	0	200	0	200	nA
L1-L24	Input current (low level)	XIII	X	0	-200	0	-200	0	-200	0	-200	0	-200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
N1-N12	Transition time, (t_{TLH}), C_L= 50 pF, V_{DD}= 5V	XV	S	20	200	20	200	22	220	15	150	24	245	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N13-N24	Transition time, (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	XV	Q	68	680	68	680	75	750	51	515	83	835	ns
P1-P12	Transition time, (t_{THL}), C_L = 50 pF, V_{DD} = 5V	XVI	S	20	200	20	200	22	220	15	150	24	245	ns
P13-P24	Transition time(t_{THL}), C_L = 150 pF, V_{DD} = 5V	XVI	Q	68	680	68	680	75	750	51	515	83	835	ns
Q1-Q24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	XVII	S	25	250	25	250	27	275	19	190	30	305	ns
Q25-Q48	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	XVII	Q	45	450	45	450	50	500	34	340	55	550	ns
R1-R24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	XVIII	S	25	250	25	250	27	275	19	190	30	305	ns
R25-R48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	XVIII	Q	45	450	45	450	50	500	34	340	55	550	ns
S1-S12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	XIX	Q	10	100	10	100	11	110	7	75	12	125	ns
T1-T12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	XX	Q	10	100	10	100	11	110	7	75	12	125	ns
U1-U24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	XXI	Q	10	100	10	100	11	110	7	75	12	125	ns
V1-V24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	XXII	Q	10	100	10	100	11	110	7	75	12	125	ns
W1-W12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	XXIII	Q	8	80	8	80	9	90	6	60	10	100	ns
X1-X12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	XXIV	Q	8	80	8	80	9	90	6	60	10	100	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	XXV	Q	8	80	8	80	9	90	6	60	10	100	ns
Z1-Z24	Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	XXVI	Q	8	80	8	80	9	90	6	60	10	100	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1,BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J and L through CC shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions K and L shall be performed at 25°, 60°, and 100°C only using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.1.2 Transient thermal test. For transient thermal test, use the same test limits as the maximum class temperature conditions.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I	Class II	Unit
Quiescent power dissipation	1.8 max	1.8 max	mW
Input voltage			
Low level V_{DD} = 5V	0.0 min, 1.5 max	0.0 min, 1.5 max	V
V_{DD} = 15V	0 min, 4 max	0 min, 4 max	V
High level V_{DD} = 5V	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
V_{DD} = 15V	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current (V_{DD} = 18V)			
Low level (V_{IL} = 0V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5V)			
I_{OL} (V_{OL} = 0.4V) at:			
-55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
I_{OH} (V_{OH} = 4.6V) at:			
-55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0 51 min	mA
60°C	-0.44 min	-0 44 min	mA
100°C	---	-0 36 min	mA

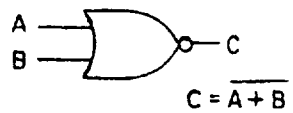
6.5 Module function. This module contains twelve independent 2 input NOR gates and two 220 pF capacitors with one side connected to V_{SS} for external use.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

INPUTS		OUTPUT
A	B	C
L	L	H
L	H	L
H	L	L
H	H	L

NOTES
1. H = High level
2. L = Low level



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

6.5.2 Paralleling of outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 1, 2, 3, and 4.

b. 5, 6, 7, and 8.

c. 9, 10, 11, and 12.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply V_{DD} is off.

d. All input signals and conditions and voltages must be removed before V_{DD} is disconnected.

e. Input voltages greater than the supply (V_{DD}-V_{SS}) should not be applied as this will forward bias the input protection diodes.

f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.7 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{Total} = (CV_{DD}^2 f)N + \text{quiescent power dissipation.}$$

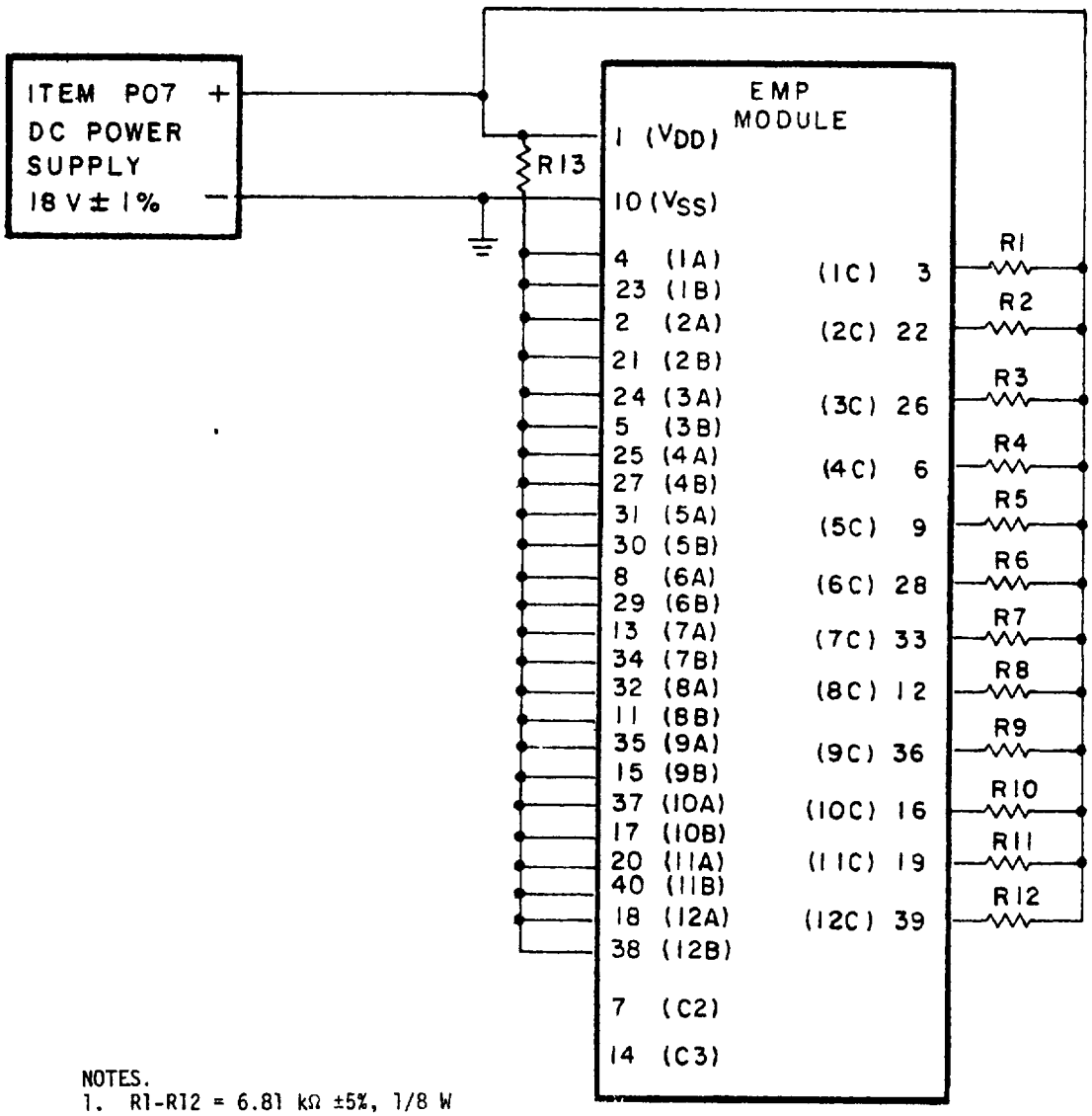
Where C = Load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

Agent
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N227)

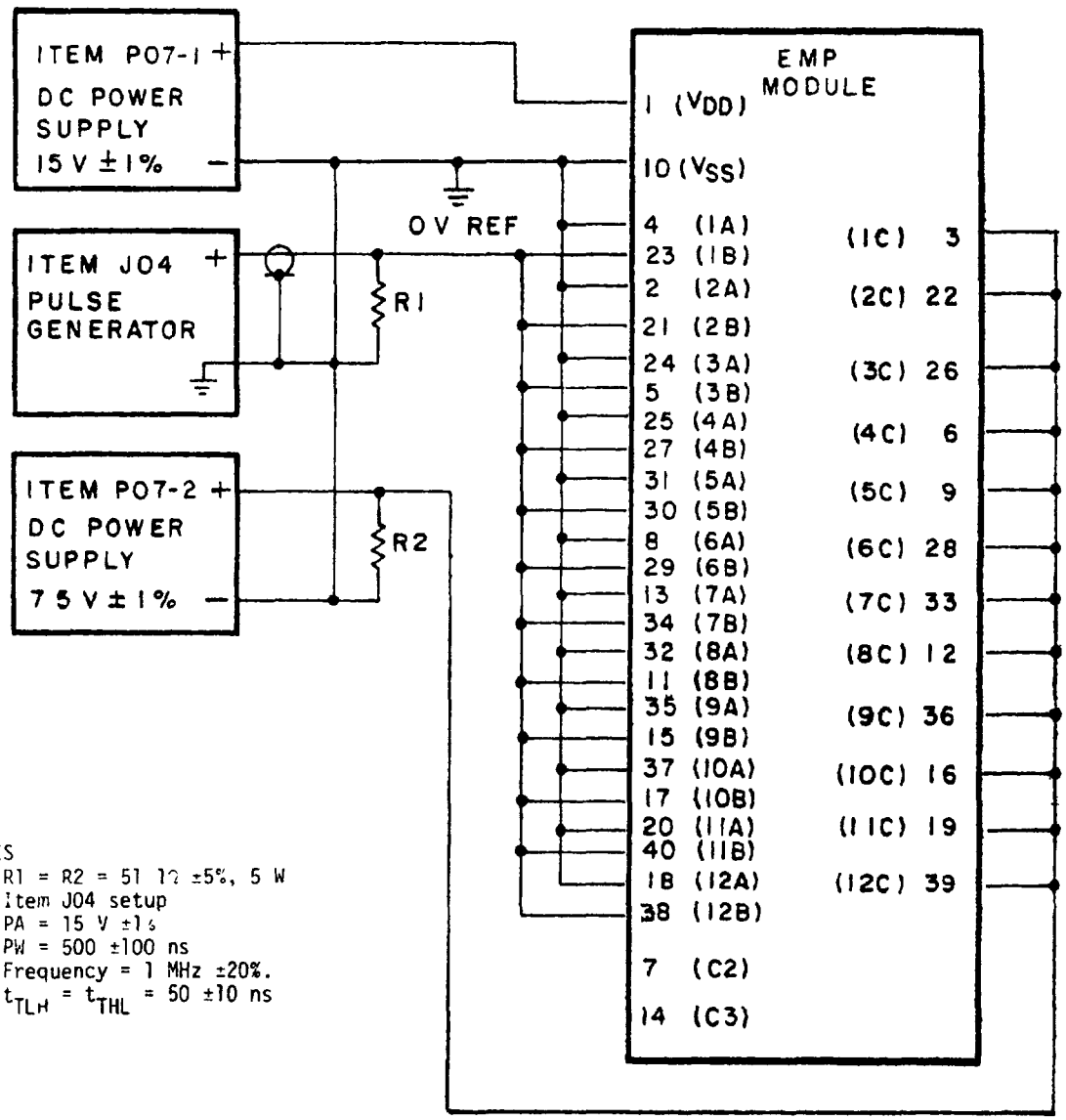
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES.
1. R1-R12 = 6.81 kΩ ±5%, 1/8 W
2. R13 = 1.5 kΩ ±5%, 1/4 W

FIGURE 2 Operating temperature and life test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = R2 = 51 Ω ±5%, 5 W
2. Item J04 setup
   PA = 15 V ±1%
   PW = 500 ±100 ns
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL} = 50 \pm 10$ ns

FIGURE 3 Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

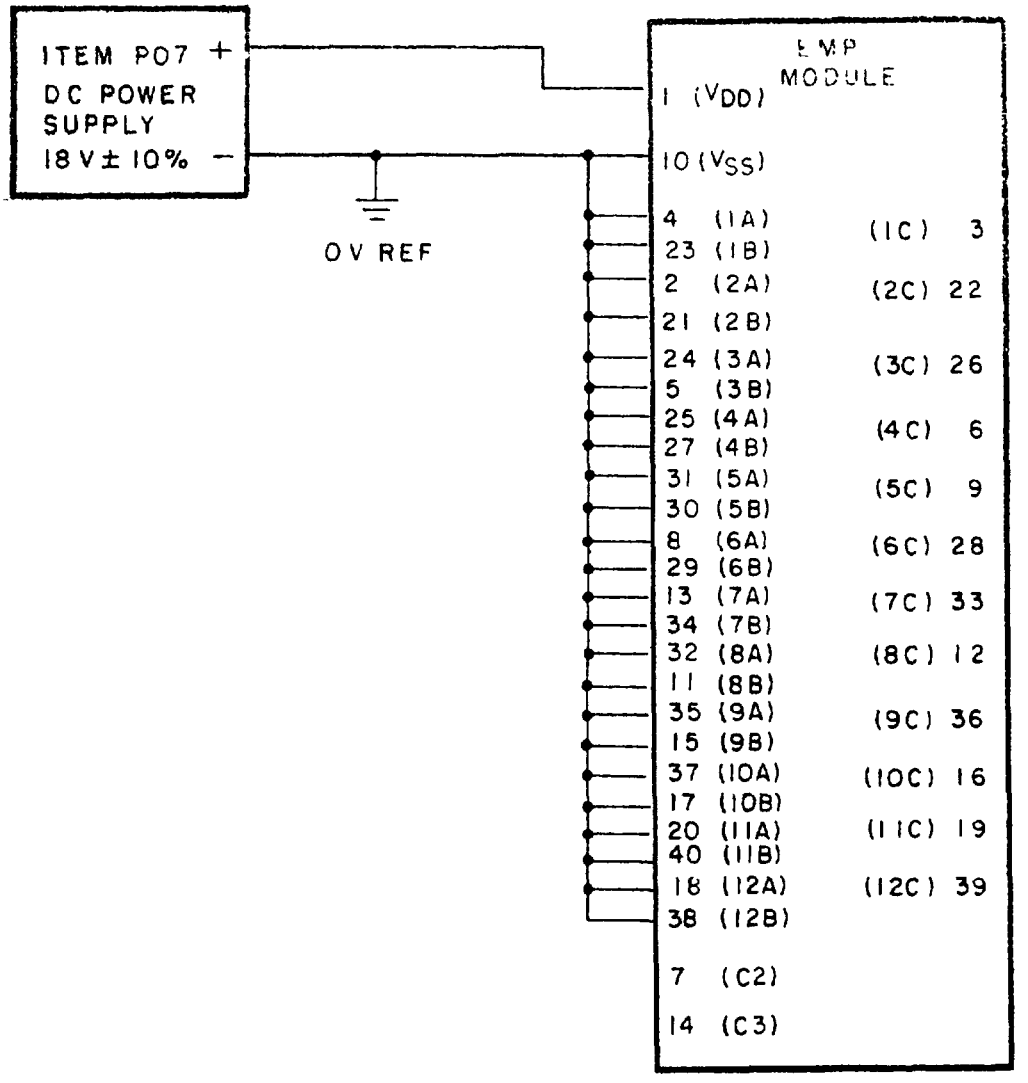


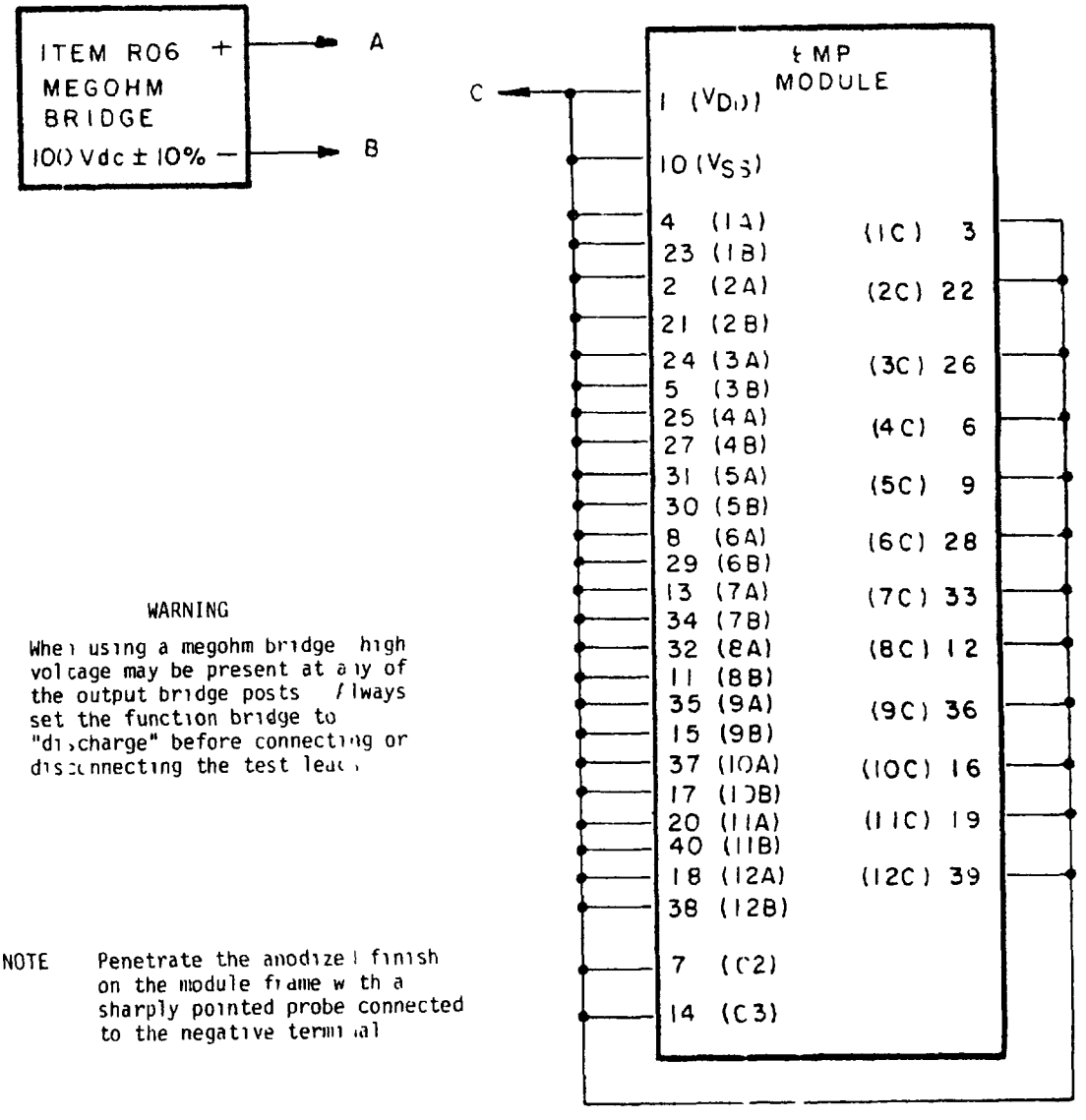
FIGURE 4 Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge high voltage may be present at any of the output bridge posts. Always set the function bridge to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE IV. Input clamp diode voltage (negative) test.

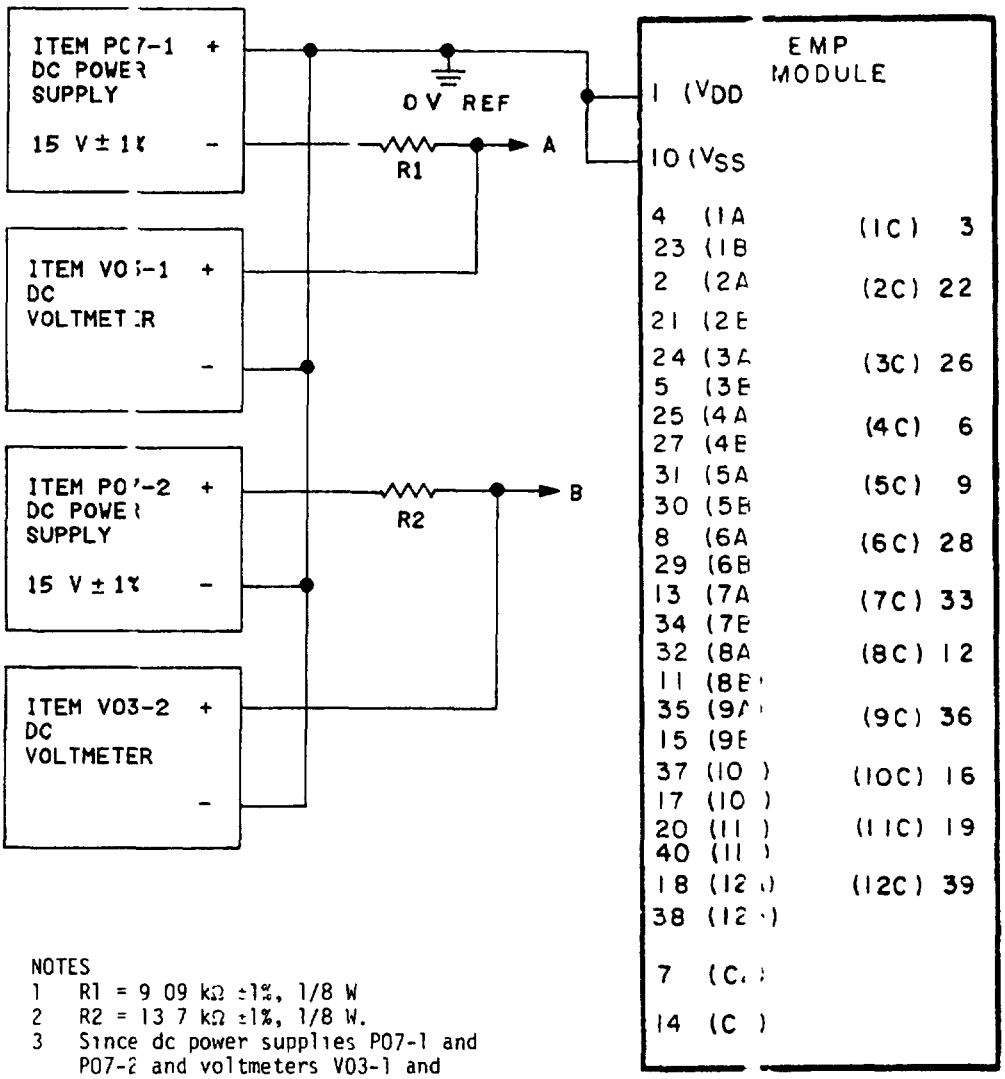
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage (negative)	4	Open	-0.4	-6.0	V	6
B2	1B	23					
B3	2A	2					
B4	2B	21					
B5	3A	24					
B6	3B	5					
B7	4A	25					
B8	4B	27					
B9	5A	31					
B10	5B	30					
B11	6A	8					
B12	6B	29					
B13	7A	13					
B14	7B	34					
B15	8A	32					
B16	8B	11					
B17	9A	35					
B18	9B	15					
B19	10A	37					
B20	10B	17					
B21	11A	20					
B22	11B	40					
B23	12A	18					
B24	12B	38					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage (positive)	Open	4	0.4	1.5	V	6
C2	1B		23				
C3	2A		2				
C4	2B		21				
C5	3A		24				
C6	3B		5				
C7	4A		25				
C8	4B		27				
C9	5A		31				
C10	5B		30				
C11	6A		8				
C12	6B		29				
C13	7A		13				
C14	7B		34				
C15	8A		32				
C16	8B		11				
C17	9A		35				
C18	9B		15				
C19	10A		37				
C20	10B		17				
C21	11A		20				
C22	11B		40				
C23	12A		18				
C24	12B		38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = 9 09 kΩ ±1%, 1/8 W
2 R2 = 13 7 kΩ ±1%, 1/8 W.
3 Since dc power supplies PO7-1 and PO7-2 and voltmeters VO3-1 and VO3-2 are not used simultaneously, only one dc power supply PO7 and one voltmeter VO3 is required

FIGURE 6 Input clamp diode voltage (negative and positive) test circuit for tables IV and V

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level, I_{OH} = 0) test.

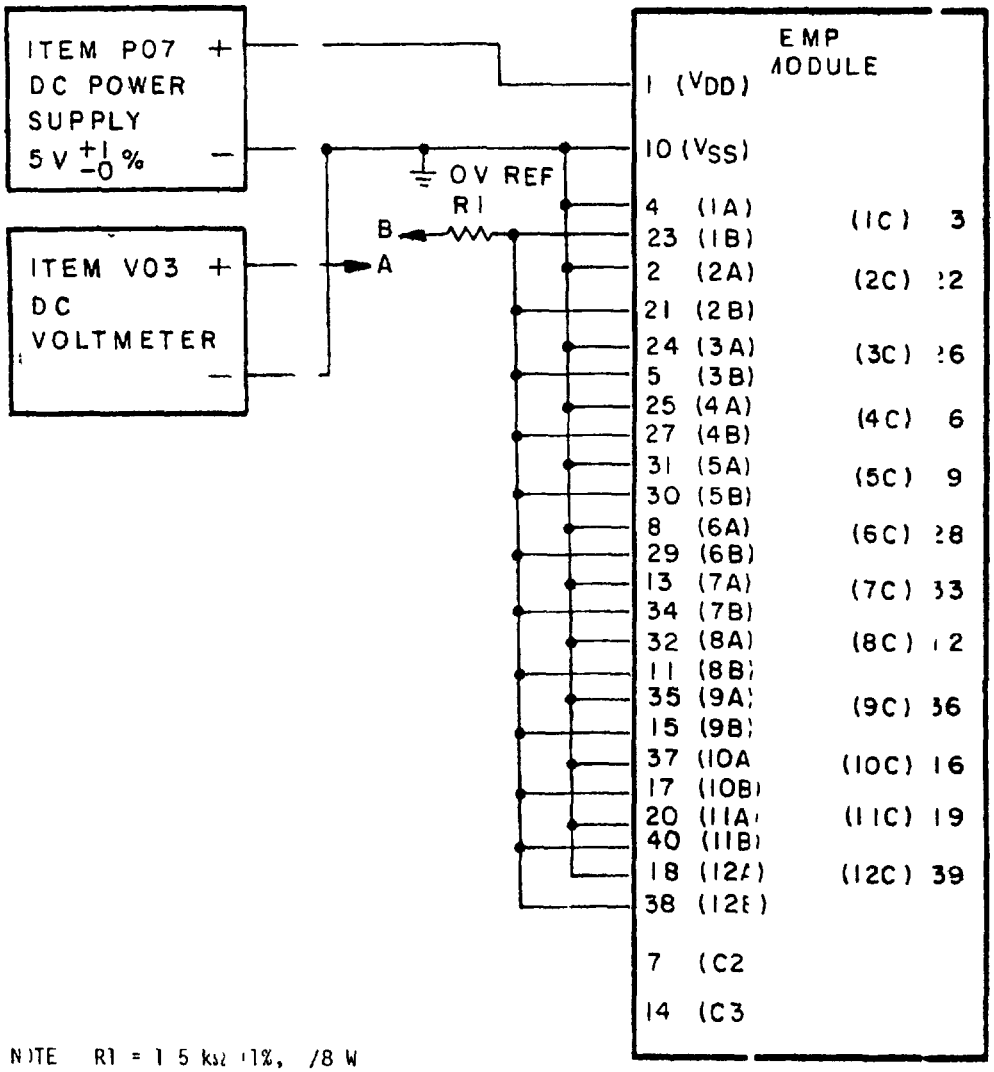
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1C Output voltage (high-level), I_{OH} = 0	3	10	4.95	5.05	V	7
D2	2C	22					
D3	3C	26					
D4	4C	6					
D5	5C	9					
D6	6C	28					
D7	7C	33					
D8	8C	12					
D9	9C	36					
D10	10C	16					
D11	11C	19					
D12	12C	39					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage (low-level, I_{OL} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1C Output voltage (low-level), I_{OL} = 0	3	1	0.00	0.05	V	7
E2	2C	22					
F3	3C	26					
E4	4C	6					
E5	5C	9					
E6	6C	28					
E7	7C	33					
F8	8C	12					
E9	9C	36					
E10	10C	16					
E11	11C	19					
E12	12C	39					

Source: https://assist.dla.mil Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTE R1 = 1.5 kΩ ±1%, /8 W

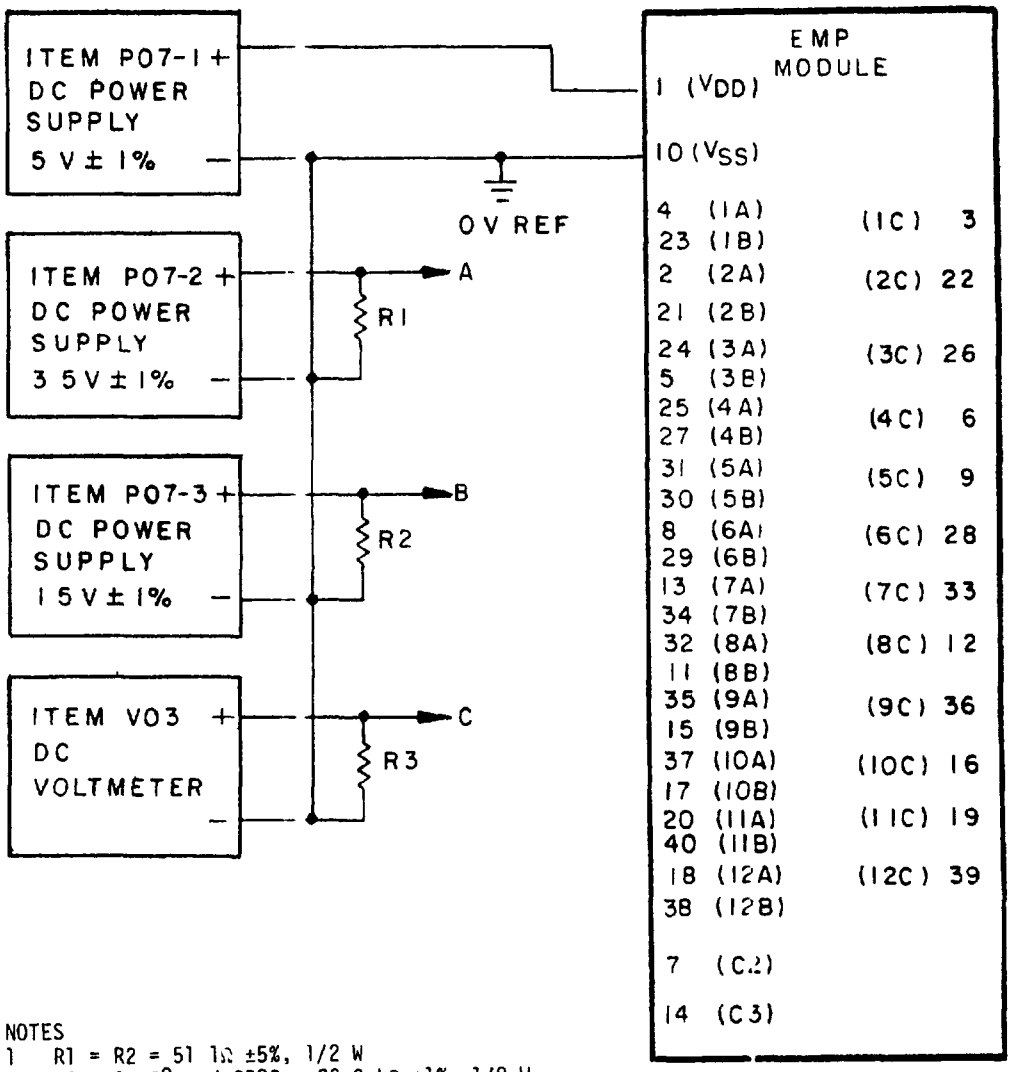
FIGURE 7 Output voltage (high-level, I_{OH} = 0 and low-level, I_{OL} = 0) test circuit for tables V and VII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage (high-level, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1C Output voltage (high level), V_{DD} = 5V	2,21,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18	4,23	3	4.6	5.0	V	8
F2	2C	4,23,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18	2,21	22				
F3	3C	4,23,2,21,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18	24,5	26				
F4	4C	4,23,2,21,24,5,31,30,8,29, 13,34,11,32,35,15,37,17, 20,40,38,18	25,27	6				
F5	5C	4,23,2,21,24,5,25,27,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18	31,30	9				
F6	6C	4,23,2,21,24,5,25,27,31,30, 13,34,11,32,35,15,37,17,20, 40,38,18	8,29	28				
F7	7C	4,23,2,21,24,5,25,27,31,30, 8,29,11,32,35,15,37,17,20, 40,38,18	13,34	33				
F8	8C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,35,15,37,17,20, 40,38,18	32,11	12				
F9	9C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,37,17,20, 40,38,18	35,15	36				
F10	10C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,20, 40,38,18	37,17	16				
F11	11C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,37, 17,38,18	20,40	19				
F12	12C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,37, 17,20,40	18,38	39				

Source: https://assist.dla.mil 21 Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = R2 = 51 1Ω ±5%, 1/2 W
2 R3 = At 0° and 25°C 23 2 kΩ ±1%, 1/8 W.
= At 60°C - 27 4 !Ω ±1%, 1/8 W.
= At -55°C - 18.2 kΩ ±1%, 1/8 W
= At 100°C - 32 4 kΩ ±1%, 1/8 W

FIGURE 8 Output voli ge (high-level, V_{DD} = 5 V) test circuit for table VIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage (low-level, V_{DD} = 5V) test.

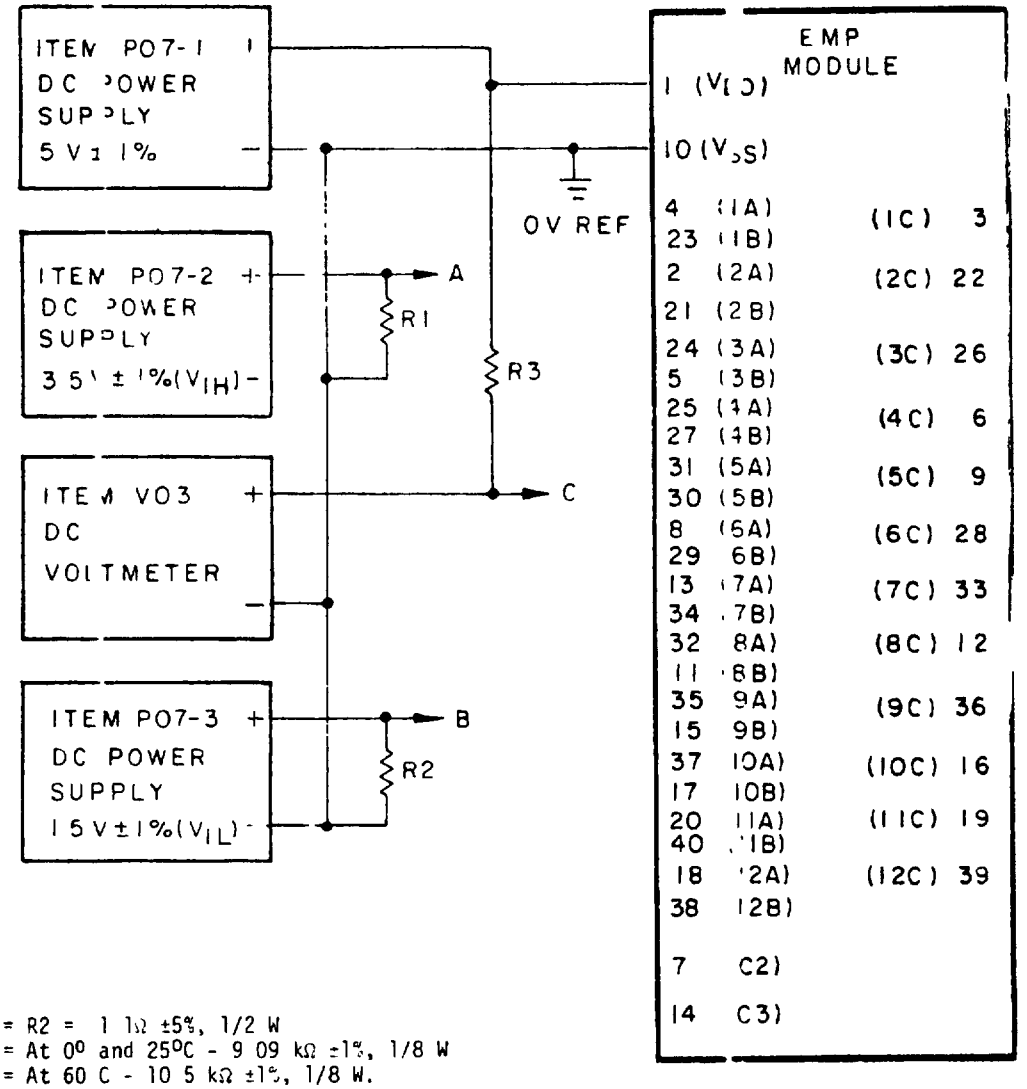
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1C Output voltage (low level), V_{DD} = 5V	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	3	0.0	0.4	V	9
G2	1C	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	3				
G3	2C	2	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	22				
G4	2C	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	22				
G5	3C	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	26				
G6	3C	5	4,23,2,21,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	26				
G7	4C	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	6				
G8	4C	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	6				
G9	5C	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	9				
G10	5C	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	9				
G11	6C	8	4,23,2,21,24,5,25,27,31,30,29,13, 34,11,32,35,15,37,17,20,40,38,18	28				
G12	6C	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18	28				
G13	7C	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18	33				
G14	7C	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18	33				
G15	8C	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18	12				
G16	8C	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18	12				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage (low-level, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G17	9C Output voltage (low level), V_{DD} = 5V	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,38,18	36	0.0	0.4	V	9
G18	9C	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18	36				
G19	10C	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18	16				
G20	10C	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18	16				
G21	11C	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18	19				
G22	11C	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18	19				
G23	12C	18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38	39				
G24	12C	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = R2 = 1.1Ω ±5%, 1/2 W
2 R3 = At 0° and 25°C - 9.09 kΩ ±1%, 1/8 W
= At 60 C - 10.5 kΩ ±1%, 1/8 W.
= At -5 °C - 7.15 kΩ ±1%, 1/8 W
= At 10· °C - 12.7 kΩ ±1%, 1/8 W

FIGURE 9 Output voltage (low-level, V_{DD} = 5 V) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage (high-level, V_{DD} = 15V) test.

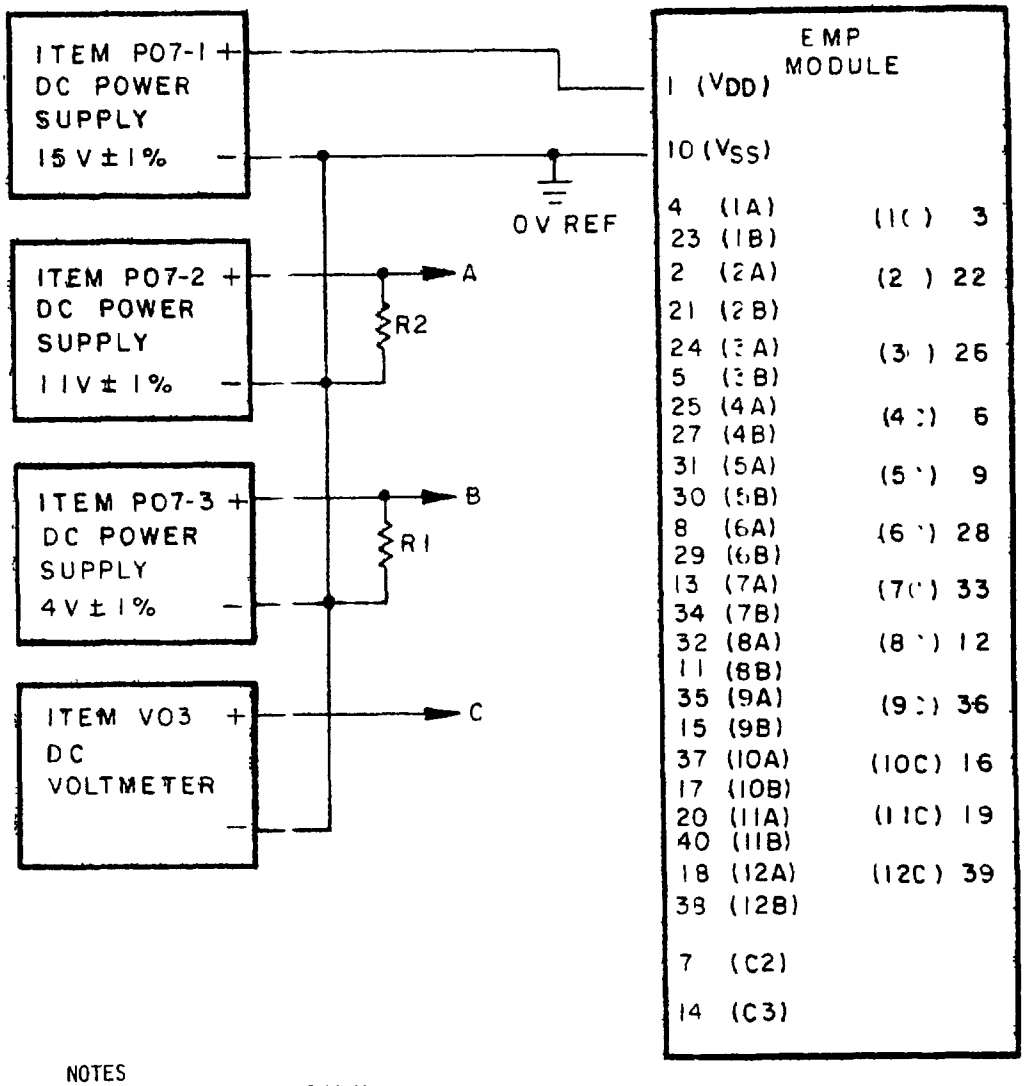
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C Output voltage (high level), V_{DD} = 15V	Open	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20,40, 38,18	3	13.5	15.0	V	10
H2	2C			22				
H3	3C			26				
H4	4C			6				
H5	5C			9				
H6	6C			28				
H7	7C			33				
H8	8C			12				
H9	9C			36				
H10	10C			16				
H11	11C			19				
H12	12C			39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage (low-level, V_{DD} = 15V) test.

Test condition.	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1C Output voltage (low level), V_{DD} = 15V	4,2,24,25,31,8,13,32, 35,37,20,18	23,21,5,27,30,29, 34,11,15,17,40,38	3	0.0	1.5	V	10
J2	2C			22				
J3	3C			26				
J4	4C			6				
J5	5C			9				
J6	6C			28				
J7	7C			33				
J8	8C			12				
J9	9C			36				
J10	10C			16				
J11	11C			19				
J12	12C			39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 51 1Ω ±5% 1/2 W
2. R2 = 576Ω ±5%, 1/4 W

FIGURE 10. Output voltage (high-level and low-level, V_{DD} = 15 V) test circuit for tables X and XI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1A Input current (high level)	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	0	200	nA	11
K2	1B	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K3	2A	2	4,23,21,24,5,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
K4	2B	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K5	3A	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K6	3B	5	4,23,2,21,24,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
K7	4A	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K8	4B	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K9	5A	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K10	5B	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K11	6A	8	4,23,2,21,24,5,25,27,31,30,29,13 34,11,32,35,15,37,17,20,40,38,18				
K12	6B	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18				
K13	7A	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18				
K14	7B	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18				
K15	8A	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18				
K16	8B	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K17	9A Input current (high level)	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,38,18	0	200	nA	11
K18	9B	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18				
K19	10A	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18				
K20	10B	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18				
K21	11A	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18				
K22	11B	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18				
K23	12A	18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38				
K24	12B	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

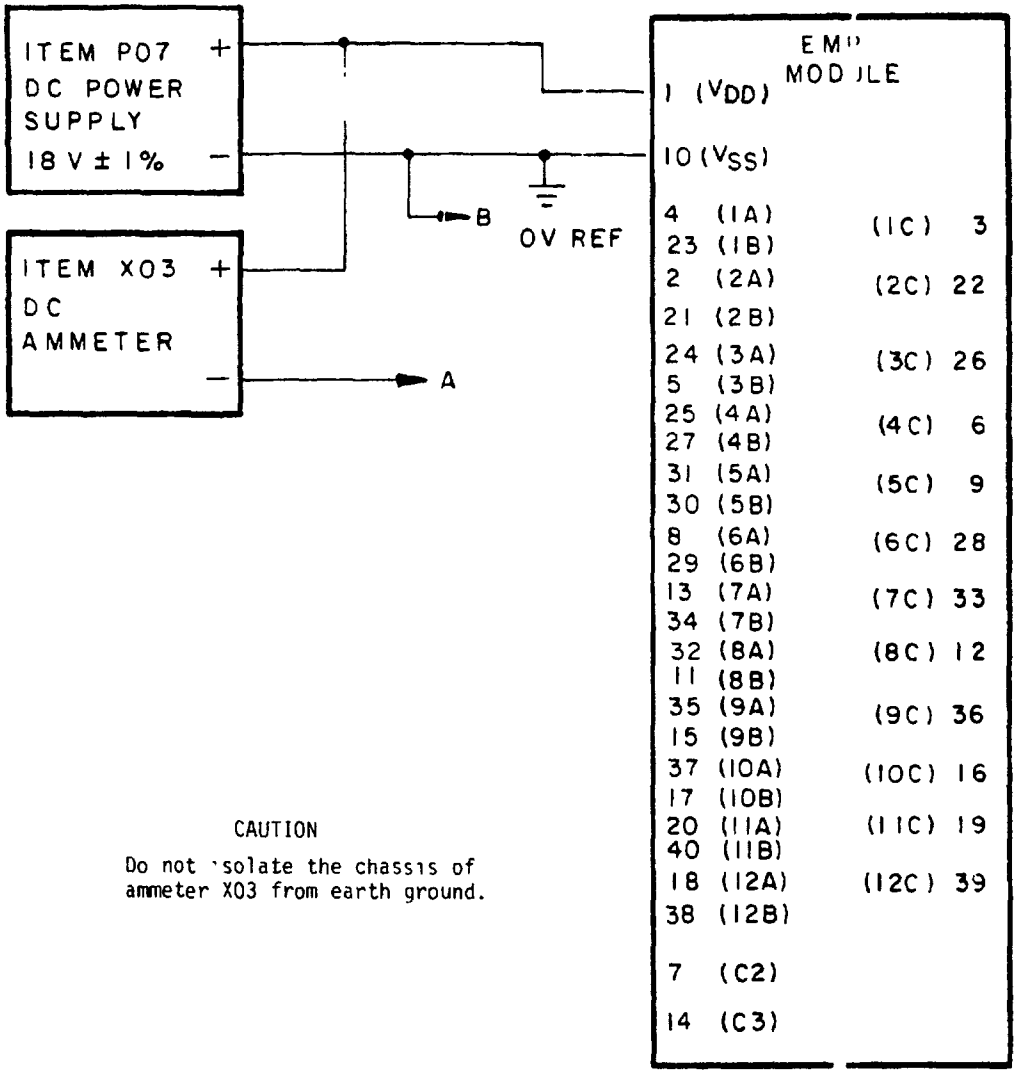


FIGURE 11 Input current (high-level) test circuit for table XII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (low-level) test.

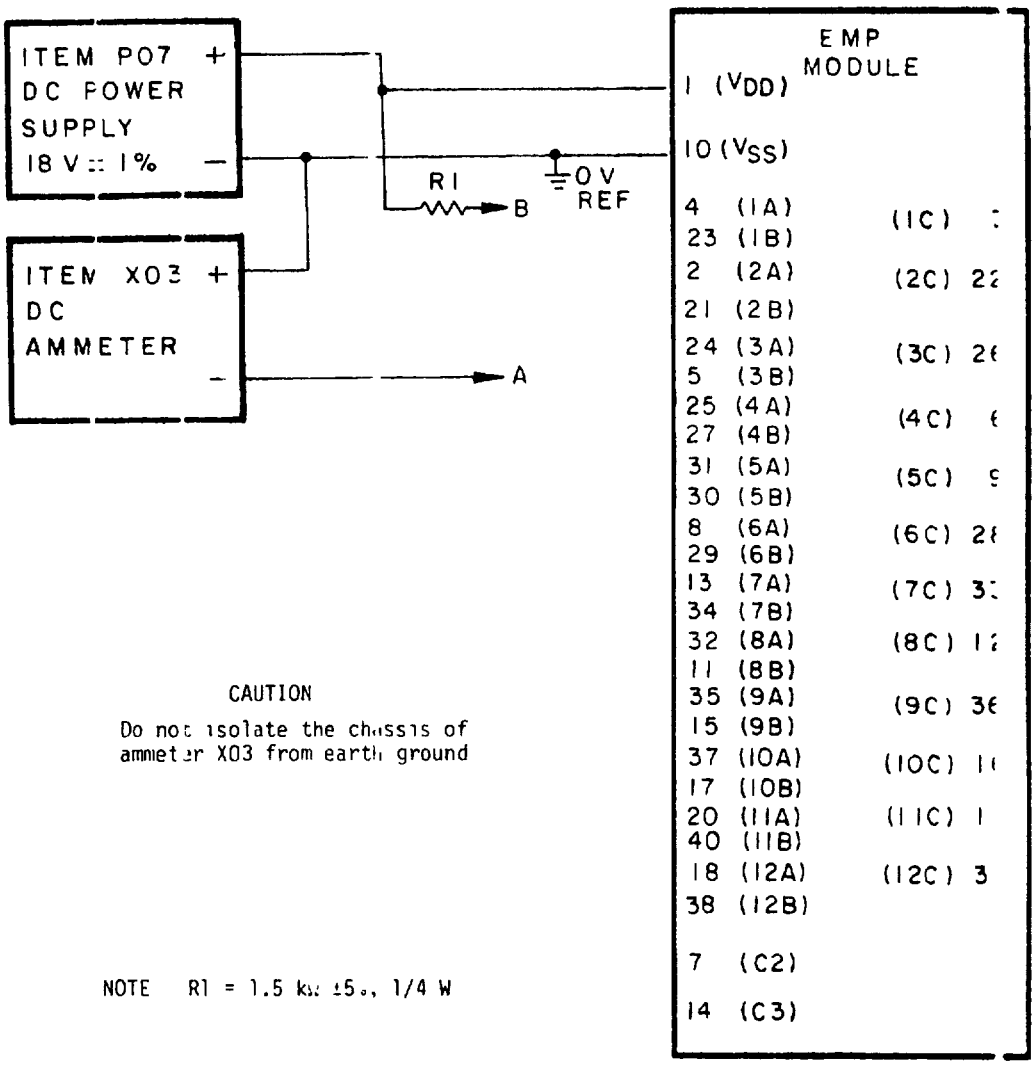
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1A Input current (low level)	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	0	-200	nA	12
L2	1B	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L3	2A	2	4,23,21,24,5,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
L4	2B	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L5	3A	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L6	3B	5	4,23,2,21,24,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
L7	4A	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L8	4B	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L9	5A	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L10	5B	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,36,16				
L11	6A	8	4,23,2,21,24,5,25,27,31,30,29,13 34,11,32,35,15,37,17,20,40,38,18				
L12	6B	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18				
L13	7A	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18				
L14	7B	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18				
L15	8A	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18				
L16	8B	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L17	9A Input current (low level)	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,38,18	0	-200	nA	12
L18	9B	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18				
L19	10A	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18				
L20	10B	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18				
L21	11A	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18				
L22	11B	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18				
L23	12A	18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38				
L24	12B	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X03 from earth ground

NOTE R1 = 1.5 kΩ ±5%, 1/4 W

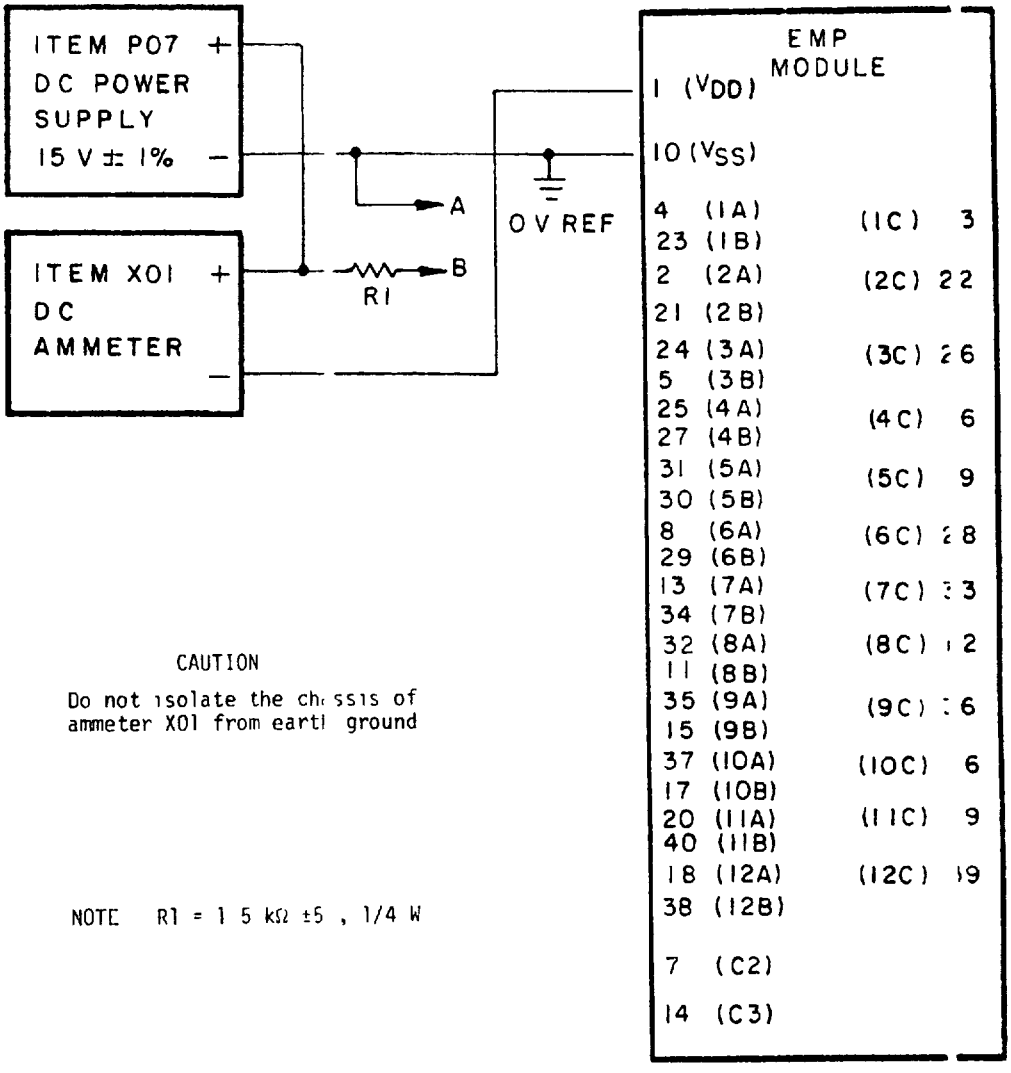
FIGURE 12. Input current (low-level) test circuit for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	Open	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18	0	6	µA	13
M2	Power supply current, high level	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18	Open	0	6	µA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTE R1 = 1.5 kΩ ±5 , 1/4 W

FIGURE 13 Power supply current (low-level and high-level) test circuit for table XIV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1C Transition time (t_{TLH}), (C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	20	200	ns	14,15
N2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
N3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
N4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
N5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
N6	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
N7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
N8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
N9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
N10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
N11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,17,38,18	20	19				
N12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLF XV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N13	1C Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	68	680	ns	14,15
N14	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
N15	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
N16	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
N17	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
N18	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
N19	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
N20	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
N21	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
N22	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
N23	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
N24	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVI. Transition time (t_{THI}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	1C Transition time (t_{THI}), C_l = 50 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	20	200	ns	14,15
P2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
P3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
P4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
P5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
P6	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
P7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
P8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
P9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
P10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
P11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,38,18	20	19				
P12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P13	1C Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	68	680	ns	14,15
P14	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
P15	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
P16	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
P17	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
P18	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
P19	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
P20	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
P21	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
P22	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
P23	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
P24	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	25	250	ns	14,15
Q2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Q3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Q4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Q5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Q6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Q7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Q8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Q9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Q10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Q11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Q12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVII. Transition time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q13	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	25	250	ns	14,15
Q14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Q15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Q16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Q17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Q18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Q19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Q20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Q21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Q22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Q23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Q24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q25	1A to 1C Propagation delay time, (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	45	450	ns	14,15
Q26	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Q27	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Q28	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Q29	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Q30	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Q31	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Q32	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Q33	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Q34	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Q35	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Q36	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVII. Transition time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q37	1B to 1C Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	45	450	ns	14,15
Q38	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Q39	3B to 3C	4,23,21,2,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Q40	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Q41	5B to 5C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Q42	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Q43	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Q44	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Q45	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Q46	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Q47	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Q48	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1A to 1C Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	25	250	ns	14,15
R2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
R3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
R4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
R5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
R6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
R7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
R8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
R9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
R10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
R11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
R12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Transition time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	25	250	ns	14,15
R14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
R15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
R16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
R17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
R18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
R19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
R20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
R21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
R22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
R23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
R24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

MIL-PRF-38535 Appendix A

Test condition	Parameter	Connect A to pin	B to pin	C to pin	Limits 25 ±5°C ambient Min	Limits 25 ±5°C ambient Max	Unit	Test figure
R25	1A to 1C Propagation delay time, (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	45	450	ns	14,15
R26	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
R27	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
R28	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
R29	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
R30	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17, 0,40,38,18	8	28				
R31	7A to 7C	4,23, 1,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
R32	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
R33	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,6, 34,13,32,11,15,37,17,20,40,38,18	35	36				
R34	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
R35	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
R36	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20 17:42Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Transition time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R37	1B to 1C Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	45	450	ns	14,15
R38	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
R39	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
R40	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
R41	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
R42	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
R43	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
R44	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
R45	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
R46	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
R47	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
R48	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	36	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XIX. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1C Transiton time (t_{TLH}), C_L = 50 pf, V_{DD} = 10V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
S2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
S3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
S4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
S5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
S6	6C	4,23,21,2,5,24,27,25,30,11,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
S7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
S8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
S9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	33	11				
S10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
S11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
S12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XX. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1C Transiton time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
T2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
T3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
T4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
T5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
T6	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
T7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
T8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
T9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
T10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
T11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
T12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PLH}) test

Test condit	Parameter	Connect			Initial 1 25°±5° ambien		Unit	Test figure
		A to pin	B to pin	C to pin	Min			
U1	A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	23,[illegible],21,24,5,25,27,31,30,8,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	4	3	10		ns	14,15
U2	[illegible]A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	2	22				
U3	A to 3C	4,23,[illegible],1,2,5,25,27,31,30,8,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	24	26				
U4	[illegible]A to 4C	4,23,[illegible],1,2,5,24,27,31,30,8,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	25	6				
U5	A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	31	9				
U6	A to 6C	4,2[illegible],1,2,5,24,27,25,30,31,29,13, 34,[illegible],32,35,15,37,17,20,40,38,18	8	28				
U7	[illegible]A to 7C	4,2[illegible],1,2,5,24,27,25,30,31,29,8, 34,[illegible],32,35,15,37,17,20,40,38,18	13	33				
U8	[illegible]A to 8C	4,23,21 2,5,24,27,25,30,31,29,8, 34,[illegible],11,35,15,37,17,20,40,38,18	32	12				
U9	[illegible] to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
U10	[illegible]A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,[illegible],32,11,15,35,17,20,40,38,18	37	16				
U11	[illegible]A to 11C	4,2[illegible],21,2,5,24,27,25,30,31,29,8, 34,[illegible],32,11,15,35,17,37,40,38,18	20	19				
U12	2A to 12C	4,2[illegible],1,2,5,24,27,25,30,31,29,8, 34,[illegible],32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXI. Transition time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U13	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	10	100	ns	14,15
U14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
U15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
U16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
U17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
U18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
U19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
U20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
U21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
U22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
U23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
U24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	1A to 1C Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
V2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
V3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
V4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
V5	5A to 5C	4,23,21,2,5,24,27,25,30,8 29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
V6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
V7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
V8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
V9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
V10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
V11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
V12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXII. Transition time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	10	100	ns	14,15
V14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	[illegible]				
V15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
V16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27					
V17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	0				
V18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
V19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
V20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	1				
V21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	16				
V22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	6				
V23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	9				
V24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1C Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
W2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
W3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
W4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
W5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
W6	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
W7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
W8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
W9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
W10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
W11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
W12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1C Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
X2	2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
X3	3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
X4	4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
X5	5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
X6	6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
X7	7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
X8	8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
X9	9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
X10	10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
X11	11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
X12	12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y1	1A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pf, V_{DD} = 15V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
Y2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Y3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Y4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Y5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Y6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Y7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Y8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Y9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Y10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Y11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Y12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y13	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	8	80	ns	14,15
Y14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Y15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Y16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Y17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Y18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Y19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	34	33				
Y20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Y21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Y22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Y23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Y24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Propagation delay time (t_{PHL}) test.

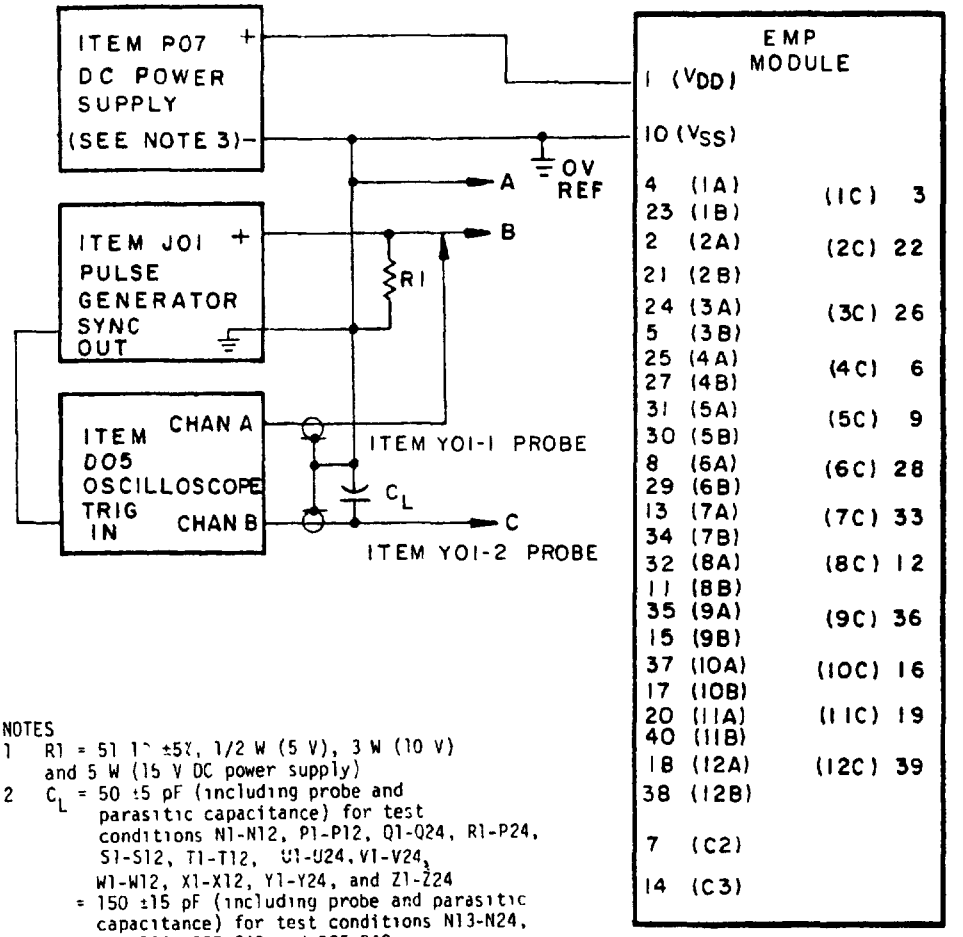
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	1A to 1C Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
Z2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Z3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Z4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Z5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Z6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Z7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Z8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Z9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Z10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Z11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Z12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	8	80	ns	14,15
Z14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Z15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Z16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Z17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Z18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Z19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Z20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Z21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Z22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Z23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Z24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

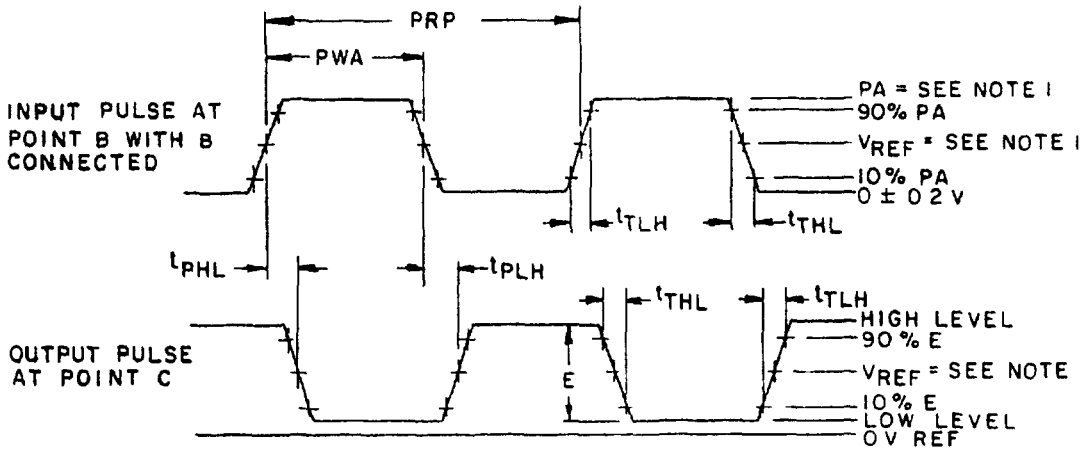
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = 51 1[illegible] ±5%, 1/2 W (5 V), 3 W (10 V) and 5 W (15 V DC power supply)
2 C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N12, P1-P12, Q1-Q24, R1-P24, S1-S12, T1-T12, U1-U24, V1-V24, W1-W12, X1-X12, Y1-Y24, and Z1-Z24
= 150 ±15 pF (including probe and parasitic capacitance) for test conditions N13-N24, P13-P24, Q25-Q48 and R25-R48
3 Set item P07 as follows
5 V 1[illegible] for test conditions N1-N24, P1-P24, Q1-Q48, and R1-R48
10 V ±1% for test conditions S1-S12, T1-T12, U1-U24, and V1-V24
15 V ±1% for test conditions W1-W12, X1-X12, Y1-Y24, and Z1-Z24

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV through XXVI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES

1 Input pulse parameters
PA = 5 0 ±0 1 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48
= 10 0 ±0 1 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24
= 15 0 ±0 1 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24
V_{REF} = 2 5 ±0 1 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48
= 5 0 ±0 1 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24
= 7 5 ±0 1 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns
PRP = 5 0 ±0.5 µs
PWA = 2 5 ±0 2 µs for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48
= 1 0 ±0 1 µs for test conditions S1-S12, T1-T12, U1-U24, V1-V24, W1-W12, X1-X12, Y1-Y24 and Z1-Z24

2 Before making the measurements, verify that the output pulse high level is ≥ 4 6 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48, ≥ 9 5 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24 and 13 5 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24 and the low level is ≤ 0 4 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48, ≤ 0 5 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24, ≤ 1 5 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24

3 The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1 00 inch (25 4 mm)

4 All voltages are referenced to ground

FIGURE 15 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV through XXVI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	0.007	0.013	µF	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. C2 and C3 capacitance test.

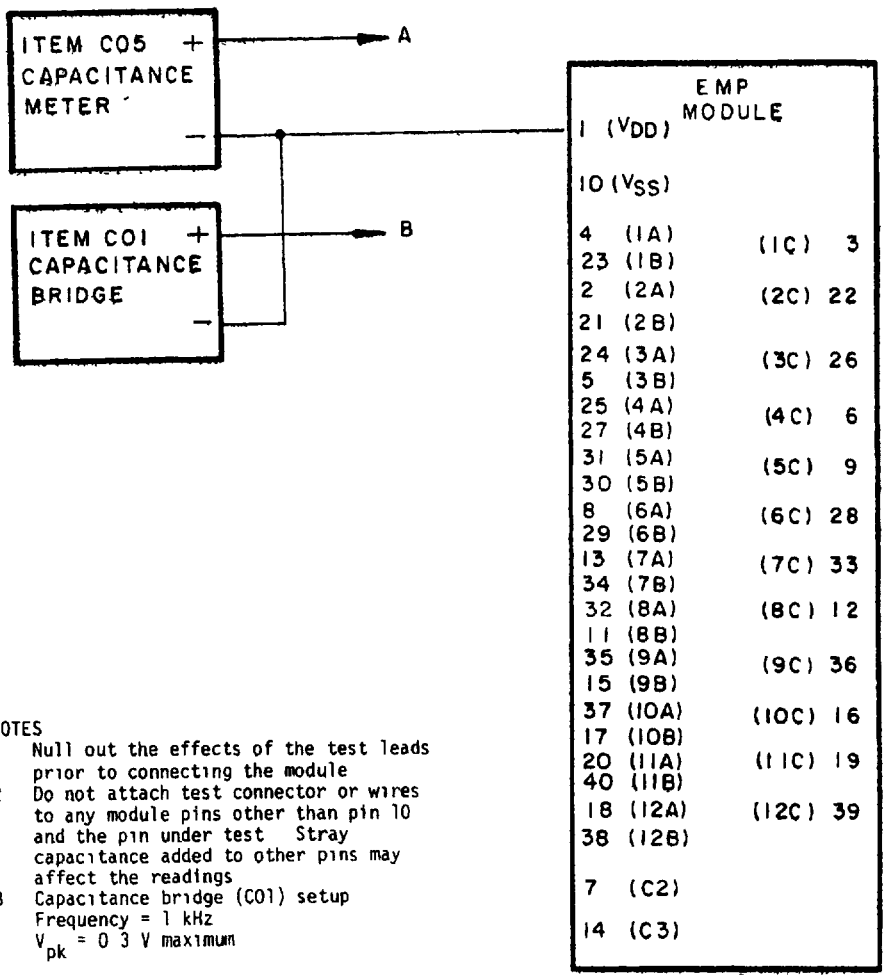
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	C2 Capacitance	7	Open	176	264	pF	16
BB2	C3 Capacitance	14	Open	176	264	pF	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

TABLE XXIX. Input capacitance test

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1A Input capacitance	4	Open	2	15	pF	16
CC2	1B	23					
CC3	2A	2					
CC4	2B	21					
CC5	3A	24					
CC6	3B	5					
CC7	4A	25					
CC8	4B	27					
CC9	5A	31					
CC10	5B	30					
CC11	6A	8					
CC12	6B	29					
CC13	7A	13					
CC14	7B	34					
CC15	8A	32					
CC16	8B	11					
CC17	9A	35					
CC18	9B	15					
CC19	10A	37					
CC20	10B	17					
CC21	11A	20					
CC22	11B	40					
CC23	12A	18					
CC24	12B	38					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.



NOTES
1 Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or wires to any module pins other than pin 10 and the pin under test Stray capacitance added to other pins may affect the readings
3 Capacitance bridge (C01) setup
Frequency = 1 kHz
V_{pk} = 0 3 V maximum

FIGURE 16 Power supply filter, C2, C3 and input capacitance test circuit for tables XXVII, XXVIII and XXIX

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-28787/263(EC) MODULES, STANDARD ELECTRONIC GATE, NOR, DIGITAL, KEY CODE EMP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Attn; ELEX 8111
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:12Z
Check the source to verify that this is the current version before use.